LAN Configuration

1. Aim

To configure a Local Area Network (LAN) by connecting multiple computers through a switch, assign IP addresses to each device, and verify communication between the devices.

2. LAN Diagram

None
```
                LOCAL AREA NETWORK (LAN)


             ┌──────────────────────┐
             │        SWITCH        │
             │       2960/24        │
             └───────┬───┬───┬───┬──┘
                     │   │   │   │
         ┌───────────┘   │   │   └───────────┐
         │               │   │               │
   ┌─────▼─────┐   ┌─────▼─────┐   ┌─────▼─────┐   ┌─────▼─────┐
   │    PC1    │   │    PC2    │   │    PC3    │   │    PC4    │
   │           │   │           │   │           │   │           │
   │192.168.1.1│   │192.168.1.2│   │192.168.1.3│   │192.168.1.4│
   └───────────┘   └───────────┘   └───────────┘   └───────────┘

                  Network: 192.168.1.0/24
                  Subnet Mask: 255.255.255.0
```

3. IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.1	255.255.255.0	192.168.1.254
PC2	192.168.1.2	255.255.255.0	192.168.1.254
PC3	192.168.1.3	255.255.255.0	192.168.1.254

PC4	192.168.1.4	255.255.255.0	192.168.1.254
Switch	192.168.1.254	255.255.255.0	—

Network Address: 192.168.1.0
Broadcast Address: 192.168.1.255
Subnet Mask: 255.255.255.0

4. Procedure

1. Connect all computers to the switch using Ethernet cables.
2. Assign a unique IP address to every computer.
3. Use the same subnet mask for all computers.
4. Configure the default gateway where required.
5. Check whether all switch ports are active.
6. Use the `ping` command to test connectivity between the computers.
7. Verify that packets are successfully transmitted between the devices.

Example:

```
None
PC1 → ping 192.168.1.2
PC1 → ping 192.168.1.3
PC1 → ping 192.168.1.4
```

Successful replies indicate that the LAN has been configured correctly.

5. Outcome

The LAN was successfully configured by connecting multiple computers through a switch. Unique IP addresses were assigned to each device, and connectivity between the devices was successfully verified using the `ping` command.

6. Challenges Faced

- Assigning unique IP addresses to every device.
- Ensuring that all devices belonged to the same network.
- Checking the correct subnet mask.
- Identifying incorrect IP configurations.
- Troubleshooting connectivity when a device did not respond to `ping`.
- Checking Ethernet cable and switch-port connections.

- Avoiding duplicate IP addresses.

7. Result

Thus, a Local Area Network was successfully implemented and configured. Communication between the connected devices was established and verified successfully.